Unsupervised Few-Shot Continual Learning for Remote Sensing Image Scene Classification

Muhammad Anwar Ma'sum, Mahardhika Pratama, *Senior Member, IEEE*, Ramasamy Savitha, *Senior Member, IEEE*, Lin Liu, Habibullah, and Ryszard Kowalczyk

***Abstract*— A continual learning (CL) model is desired for remote sensing (RS) image analysis because of varying camera parameters, spectral ranges, resolutions, etc. There exist some recent initiatives to develop CL techniques in this domain but they still depend on massive labeled samples which do not fully fit RS applications because ground truths are often obtained via field-based surveys. This article addresses this problem with a proposal of unsupervised flat-wide learning approach (UNISA) for unsupervised few-shot CL approaches of RS image scene classifications which do not depend on any labeled samples for its model updates. UNISA is developed from the idea of prototype scattering and positive sampling for learning representations while the catastrophic forgetting (CF) problem is tackled with the flat-wide learning approach combined with a ball generator to address the data scarcity problem. Our numerical study with RS image scene datasets and a hyperspectral dataset confirms the advantages of our solution. For future studies and reproductions, source codes of UNISA are shared publicly in https://github.com/anwarmaxsum/UNISA.**

***Index Terms*— Continual learning (CL), few-shot continual learning (FSCL), few-shot learning, unsupervised learning.**

I. Introduction

THE use of deep learning (DL) has been widespread in the domain of remote sensing (RS) such as Earth observation, disaster monitoring, and urban planning. This progress is mainly driven by the operations of advanced satellite technologies facilitating RS imaging technologies for convenient data collections. Nonetheless, RS domains are known to be rapidly changing because of temporal, spatial, and spectral resolution limitations, thus calling for continual learning (CL) solutions, e.g., while Earth observation sensors obtain images continuously over time, the land cover/land use classes are made available sequentially [1]. RS applications characterize four unique problems [2]: *data openness*, *temporal openness*, *spatial openness*, and *spectral openness*. Naive retraining approaches impose considerable complexities which do not scale well for continuous environments.

CL is a research area of growing interests in the RS domain where several works have been recently proposed in [1], [2], [3], [4], and [5]. The concept of model-agnostic meta-learning (MAML) [6] is implemented in [3] for CL of RS images. Lightweight incremental learning (LIL) approach is proposed in [5] for continual RS image classification. Ammour [4] presents the notion of two trainable subnetworks and [1] is based on curriculum learning. The concept of preserving network and transient network is developed in [2] for CL of RS image scene classifications. Nonetheless, these works require a lot of labeled samples which do not fit to RS applications. Labeling cost is often expensive in RS domains because ground truth is based on the field-based survey. Fig. 1 visualizes the case of CL for RS images. The number of classes cannot be fixed at once due to temporal, spatial, and spectral resolution limitations leading to the case of CL, which should learn a sequence of new target classes without the catastrophic forgetting (CF) problem.

Some attempts have been made to address the problem of labeling costs in the RS area. Zhang et al. [7] propose the concept of CNN and GCN for cross-scene hyperspectral image classifications. Zhang et al. [8] offer a single-source domain generalization framework in realm of cross-scene hyperspectral image classifications based on the supervised contrastive learning framework. Another domain generalization framework is proposed in [9] integrating linguistic modal information. Zhang et al. [10] combine the hyperspectral data and the Lidar data for collaborative land-cover classifications. Albeit these recent advances in handling the low-label situation, these works do not yet address the CL problem where a model has to cope with a sequence of different learning tasks. In addition, the problem of limited data remains an uncharted territory.

How to handle CL problems with limited samples is known as few-shot CL (FSCL) [11] and has attracted significant research interests in realm of CL. Tao et al. [11] propose the idea of neural gas, while [12] puts forward the concept of session trainable parameters. Zhang et al. [13] integrate the graph attention network to boost the performance of base task, while [14] addresses the FSCL problem using the concept of incremental vector quantization. Shi et al. [15] offer the idea of flat learning concept, and Kalla and Biswas [16] propose

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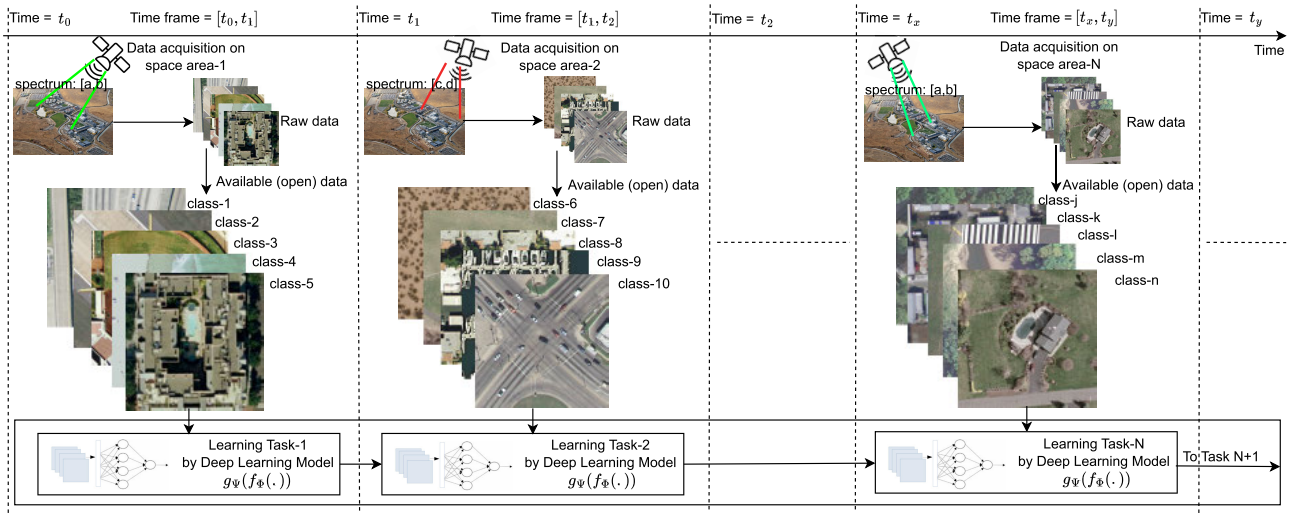


Fig. 1. Visualization of CL for RS scene classification.

the idea of stochastic classifier. All these works have not been validated with RS data. In addition, they still assume the presence of fully labeled training data albeit low sample constraint. Note that [17] does not address the CL problem although it deals with cross-domain few-shot learning problems. The motivations of this article are outlined as follows.

1) Existing works in the CL for RS data do not yet address the problem of data scarcity. That is, they suffer from the overfitting problem in the case of FSCL.
2) Although the problem of FSCL has been studied in some recent works, these approaches are not yet targeted for RS data possessing unique characteristics.
3) Existing FSCL works assume the presence of fully labeled samples, which may not hold in many practical environments. In practice, labels are scarce and expensive to obtain.

This article proposes unsupervised flat-wide learning approach (UNISA) for unsupervised FSCL (UFSCL) of RS image scene classification. UFSCL can be perceived as an extension of FSCL [11] where no labeled samples are used for model updates. Very few samples are only made available for class inferences. UNISA is developed from the concept of deep clustering network where the representation training process is driven with the concept of prototype scattering and positive sampling [18] avoiding the class collision issue while attaining uniform representation. This strategy combines the concept of contrastive learning and noncontrastive learning where the prototype scattering loss assures the uniformity of representation while the positive sampling loss overcomes the class collision issue. The CF problem is resolved with the union of flat learning concept [15] and wide learning concept [19] seeking and maintaining the flat-wide local minima regions. That is, the base learning task explores the flat-wide minima regions via noise perturbation while clamping network parameters to stay around the flat-wide local optimum regions during the few-shot learning tasks. The idea of ball generator [20] is implemented during the few-shot learning tasks as a feature-level data augmentation approach to address the problem of data scarcity.

This article contains four major contributions.

1) We propose a challenging but practical problem, unsupervised FSCL (UFSCL) problems. This problem emphasizes the low-label problem in the FSCL setting where no labeled samples are offered for model updates. We proposed unsupervised flat-wide learning (UNISA) approach as a solution of the UFSCL problem. To the best of our knowledge, the existing FSCL approaches exclude the low-label problem and thus are not directly applicable to the UFSCL setting.
2) We propose a novel UNISA based on the concept of prototype scattering and positive sampling. This approach goes one step ahead of the current flat-wide learning approach [21] only designed in the fully supervised learning contexts.
3) A new joint loss function for UFSCL problems is put forward. The representation learning component aims to extract meaningful representation without any labels and the ball data generator component is meant to enrich latent space representation while the wide learning component and the regularization term protect against the CF problem.
4) Our rigorous experiment shows our proposed method is superior to the existing state-of-the-art algorithms in six popular RS datasets and robust in various scenarios.

II. Related Works

A. Continual Learning

CL has rapidly progressed in the literature where the major challenge is to overcome the CF problem when learning new tasks, thus making possible for knowledge accumulation overtime. The existing approaches consist of three groups [22]: regularization-based approach, architecture-based approach, and memory-based approach. Regularization-based approach [19], [23], [24], [25], [26], [27], [28], [29] relies on a penalty term constraining important parameters from significant deviations to relieve the CF problem. This approach, however, does not scale well for long sequences of tasks and

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depends on the task ID. The architecture-based approach is based on a growing network structure followed by isolation of old parameters to combat the CF problem [30], [31], [32], [33], [34], [35], [36]. This approach imposes costly computational burdens while still relying on the task IDs. The memory-based approach is based on an episodic memory storing old data samples to be interleaved with new samples when learning new tasks [37], [38], [39], [40], [41], [42], [43], [44]. Although these approaches offer strong performances with the absences of task IDs, they are not feasible for the FSCL problem due to the few sample limitation. The application of episodic memory in such setting likely leads all the samples of a task to be stored in the memory, thus being against the CL spirit. CL has also attracted research attentions in the RS field as shown in [1], [2], [3], [4], [5]. These approaches require a lot of labeled data samples, and on the other hand RS data usually do not carry long historical information. Recent works on the CL problem use a frozen pretrained ViT backbone and learned additional parameters called prompts to adapt to the sequence of tasks [45], [46], [47]. The other studies enhance the prompt learning with a self-evolving structure to accommodate the increase in learned knowledge [48]. Another approach enhanced it with language models as additional information [49], [50]. Even though they are proven in the CL problem, the prompt-based approach is not yet validated in FSCL integrating a data scarcity challenge.

B. Few-Shot Continual Learning

FSCL [11] extends CL where it imposes a small sample constraint. That is, there exists a base learning task where many samples are supplied and a sequence of few-shot learning tasks where very few samples are accessible under the N-way K-shot configuration, i.e., every task carries N classes where each class only has K samples. Tao et al. [11] use the concept of neural gas, Mazumder et al. [12] put forward the idea of session trainable parameters, Zhang et al. [13] put forward the idea of graph attention network, Chen et al. [14] use the concept of incremental vector quantization, and Shi et al. [15] apply the flat minima concept. Note that the concept of flat learning in [15] does not involve the concept of wide learning and performs poorly under small base tasks because of inaccurate estimations of flat regions. In addition, Shi et al. [15] compromise the plasticity of a model when learning new concepts. Kalla and Biswas [16] combine the ideas of self-supervised learning and stochastic classifier. The FSCL problem does not address the issue of labeling costs and assumes the presence of fully labeled samples. This issue does not fit well for RS applications where the labeling cost is expensive. Recent works in FSCL try to calibrate the classes' prototypes and use them as classifiers to minimize forgetting [51], [52], [53], [54]. The other approaches adjust model weight e.g., with geometric transformation or adding subnetworks [55], [56]. Note that the methods are trained in a supervised way, both in the base task and the few-shot tasks, and require abundant labeled samples for base task training. Thus, they are not yet proven in the case of low labels in the base task.

C. Unsupervised Continual Learning

Unsupervised continual learning (UCL) [57] addresses the issue of labeling cost in CL where the training process is performed with the absence of any labeled samples. Small fractions of labels can be made available only for class inferences. Ashfahani and Pratama [34], Pratama et al. [35], and Smith et al. [58] cope with this problem via deep clustering networks with a self-evolving property. The key difference with our proposed UFSCL in this article lies in the access of a lot of unlabeled samples for UCL, whereas the UFSCL does not enjoy such access.

D. CL in RS

CL has triggered active researches in the RS domain. The concept of MAML [6] is integrated in [3] for lifelong learning of RS images. The concept of LIL is proposed in [5] for CL of RS images. Ammour [4] adopts the idea of two trainable subnetworks while [1] proposes the notion of curriculum learning. The concept of preserving and transient networks is proposed in [2]. In [59], distribution reliability assessment based on incremental learning (DRAIL) is proposed for automatic target recognition. Liu et al. [60], [61] emphasize the importance of knowledge distillation concept RS classification. A novel exemplar selection is proposed in [62] for class-incremental learning algorithm and applied to the RS data. Synthetic aperture radar automatic target recognition (SARATR) is proposed in [63] and based on the few-shot class-incremental learning algorithm. Liu et al. [64] proposed label-consistent classifier and gradual regression to boost model performance on few-shot incremental object detection in RS. Lu et al. [65] used prompting approach and dual-frequency approaches for FSCL in RS. Wang et al. [66] proposed multiscale feature representations and multiscale fusion to enhance the model discriminative capability for FSCL in RS. All these works are not compatible for the UFSCL problem proposed here because they are still designed for ordinary CL problems excluding low-label and low-data situations in the CL problems.

III. Preliminaries

A. Problem Definition

UFSCL is defined as a sequential learning problem over a sequence of tasks, $\mathcal{T}_1, \ldots, \mathcal{T}_K, k \in \{1, \ldots, K\}$ under the presence of both label and data scarcities. K stands for the number of tasks unknown in practice. Unlike conventional FSCL problem [11], a model is fed only by unlabeled samples $\mathcal{T}_k = \{x_i^k\}_{i=1}^{|\mathcal{T}_k|}$ where $|.|$ stands for the cardinality symbol and $x_i^k \in X$ denotes the ith image of the kth task. A very small subset of labels can be accessed only for class inferences $Y_k = \{y_i\}_{i=1}^{|Y_k|}$, $|Y_k| \ll |\mathcal{T}_k|$ where $y_i \in \mathcal{Y}$ denotes a class label of the ith image and Y_k stands for the label set of the kth task. All the images are drawn from the same domain $\mathcal{X} \times \mathcal{Y} \in \mathcal{D}$. We only focus on the challenging class-incremental learning setting [67] here where a model is configured under a single head structure with the absence of any task IDs. That is, each task carries different and nonoverlapping label sets $\forall_{k,(k-1)} Y_k \cap Y_{k-1} \in \emptyset$.

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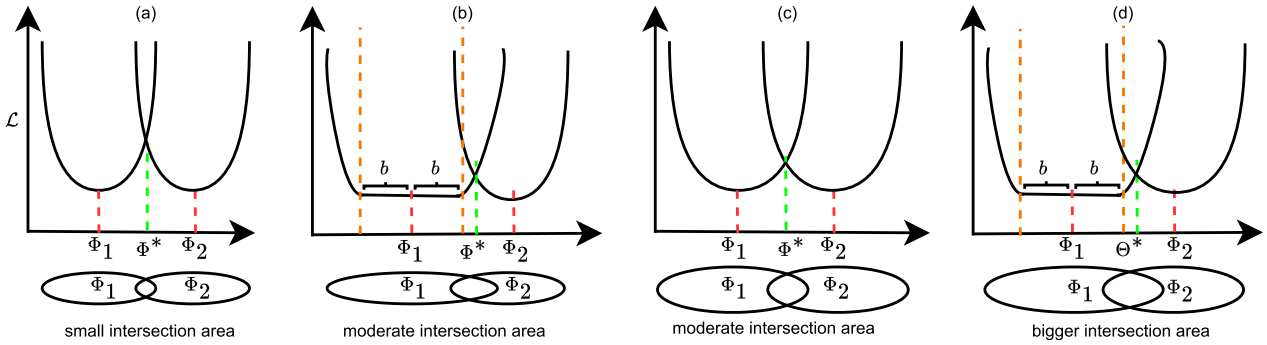


Fig. 2. Visualization of model parameters in (d) flat-wide learning compared with (a) standard learning, (b) flat minima learning, and (c) wide minima learning.

As with the FSCL problem, the UFSCL problem comprises a base learning task $\mathcal{T}_1$ and streaming few-shot learning tasks $\mathcal{T}_k$. There exist large samples of the base learning task $\mathcal{T}_1$, whereas the data scarcity issue is seen in the few-shot learning tasks $\mathcal{T}_k$. That is, the few-shot task $\mathcal{T}_k$ is configured in the N-way K-shot setting, i.e., it carries N classes where each class contains only K samples. We explore the case of $K \in [1, 5]$.

UNISA is structured as a deep clustering network composed of a feature extractor $z = f_\Phi(x) \in \Re^D : x \to z$ mapping an input image $x \in \Re^{w \times h \times c}$ to a feature space, fully connected layers $\hat{z} = g_\Psi(z) \in \Re^{\hat{D}} : z \to \hat{z}$, and a clustering module $\hat{y} = \kappa_C(\hat{z}) : \Re^{\hat{D}} \to \Delta_M$ delivering final predictions in which M denotes the output dimension.

B. Our Solution: UNISA

In this section, we introduce the main concept of our proposed method i.e., unsupervised flat-wide learning illustrated in Fig. 2. Given a DL model $g_\Psi(f_\Phi(.))$ where f and g are the feature extractor and classifier parameterized by Φ and Ψ, respectively. Standard FSCL learning optimizes Φ to recognize the learned classes from the base task (T_1) until the current task (T_2). However, since the tasks have different feature distributions, the optimal intersection of optimal parameters for T_1 and T_2 is small as shown in Fig. 2(a). Flat minima learning shown in Fig. 2(b) tries to search the flat region on the base task by perturbing Φ_1 with noise in the range [−b,b] then forced to clamp the optimal parameter Φ^* into the flat region. Wide minima learning strives to enlarge the insertion between Φ_1 and Φ_2 by scratching both the loss curves as shown in Fig. 2(c). Our proposed method, i.e., flat-wide learning, leverages both the properties. In the base task, it finds a flat region while simultaneously scratching its loss curve. In the few-shot task, the model keep applying the wide minima principle, so the intersection between Φ_1 and Φ_2 is bigger than the previous three approaches. The higher intersection area implies the model has a higher chance of recognizing classes in both the tasks.

The second uniqueness is that our proposed method is trained in an unsupervised way, while the previous three approaches are trained in a supervised way. Instead of the actual label, our approach uses a pseudolabel produced by an unsupervised learning mechanism, i.e., clustering to optimize Φ. Third, our proposed method creates multiple representations for each input. In each iteration, given an input x, our method creates representations $f_\Phi(x)$ and $f_{\Phi'}(x)$ where Φ' is an alternate version of Φ by applying a moving average operation. This enrichment of representation plays an important role in optimizing Φ. These last two principles try to answer the data scarcity challenge where only a small number of labeled data are available. The details of our proposed method are presented in Section IV.

IV. Base Learning Session $k = 1$

This section outlines the procedure of base learning session of UNISA whose goal is to solicit flat-wide local minima to be retained throughout its learning process via representation learning of the prototype scattering and positive sampling losses.

A. Representation Learning

The clustering-friendly latent space of UNISA is attained via a representation learning method minimizing the prototype scattering and positive sampling losses [18]. This concept combines the concept of contrastive and noncontrastive learning under one roof taking advantages of their strengths while eliminating their bottlenecks. Notwithstanding that the contrastive learning strategy [68] using the InfoNCE loss allows uniform representations, it suffers from the class collision issue where negative samples of the same clusters are wrongly pushed away. On the other hand, such drawback is absent in the noncontrastive learning technique but it often results in trivial solutions leading to too many empty clusters. Let x be an instance where x^+ is its augmented version via random augmentation and $\{x_1^-, \ldots, x_m^-\}$ be a set of m negative instances, the InfoNCE loss can be written as follows:

$$\mathcal{L}_{\text{INCE}} \approx \underbrace{-\frac{g_\Psi\left(f_\Phi(x)^T\right) g_\Psi\left(f_\Phi\left(x^+\right)\right)}{\tau}}_{\text{alignment}} + \underbrace{\log \sum_{i=1}^{m} \exp\left(\frac{g_\Psi\left(f_\Phi(x)^T\right) g_\Psi\left(f_\Phi\left(x^-\right)\right)}{\tau}\right)}_{\text{uniformity}} \quad (1)$$

pulling together the positive pair (x, x^+), the alignment term, while pushing away x from a set of m negative instances,

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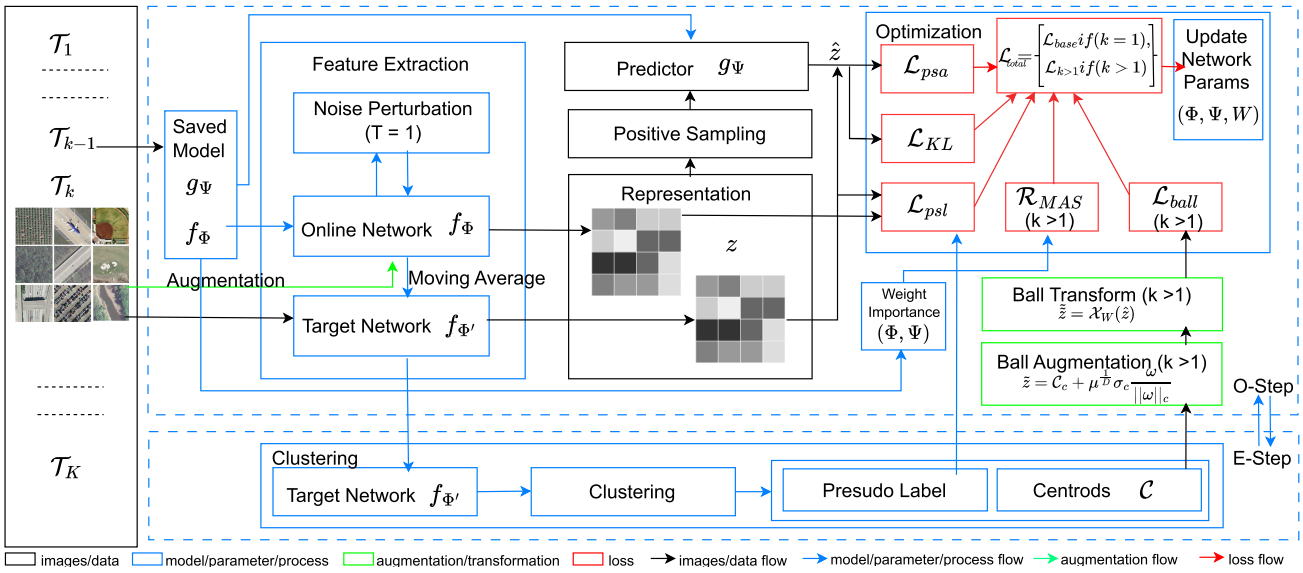


Fig. 3. Workflow of the proposed method (UNISA): UNISA is trained minimizing a joint loss function of the prototype scattering loss, the positive sampling alignment loss, the wide local minimum loss, the ball generator loss, and the regularization term. The data augmentation procedure is implemented to address the data scarcity problem.

the uniformity term. The absence of cluster representation causes the class collision issue due to the uniformity term. The uniformity term is discarded in the nonconstrastive learning technique, thus avoiding the class collision issue but suffers from nonuniform representations.

The prototype scattering loss works directly on the cluster level using the contrastive strategy. Suppose that there exist N_C clusters, one cluster is assigned as a positive cluster while the remainder $N_C - 1$ clusters are selected as negative prototypes. N_C clusters are constructed at the bottleneck layer of the fully connected module $\{C_1, C_2, \ldots, C_{N_C}\}$ and at the same time their augmented versions are also created using the random augmentation approach $\{C'_1, C'_2, \ldots, C'_{N_C}\}$. The prototype scattering loss function is formalized as follows:

$$\mathcal{L}_{\text{psl}} = \frac{1}{N_C}\sum_{c=1}^{N_C} -\frac{C_c^T C'_c}{\tau} + \frac{1}{N_C}\sum_{s=1}^{N_C}\log\sum_{\substack{c=1\\c\neq s}}^{N_C}\exp\left(\frac{C_s^T C_c}{\tau}\right). \tag{2}$$

Equation (2) guarantees uniform representations and well-separated clusters without any risk of the class collision issue because the negative instance is of different clusters.

The positive sampling alignment approach is developed to improve within-cluster compactness while further overcoming the class collision problem. This is achieved by aligning the neighboring examples with another view of a sample. Note that neighboring samples are truly positive and belong to the same class [18]. Neighboring samples themselves are constructed with the help of the Gaussian distribution controlling the influence zone of a sample to be treated as positive samples. The positive alignment loss is written as follows:

$$\mathcal{L}_{\text{psa}} = ||g_\Psi(f_\Phi(x + \epsilon\sigma)) - f'(x^+)||_2^2 \tag{3}$$

where $\epsilon \backsim \mathcal{N}(0, 1)$ and σ is a predefined constant. $f'(.)$ is a target network, momentum updated. Equation (3) should not cause any class collision issue because neighboring samples of an instance should be positive samples.

This strategy is implemented in the EM framework where the k-means clustering strategy is applied in the E step followed by the fine-tuning mechanism during the M step minimizing the proposed loss function. The target network is updated using moving average formula as [18] and [69] $\phi' = m\phi' + (1 - m)\phi$ and used in the inference phase.

B. Flat-Wide Base Learning Phase $k = 1$

The base learning phase is devised to capture flat-wide local optimum regions in which the losses are still minimized while coping with the CF problems. The flat learning concept itself [15] does not suffice because it risks on too narrow local optimum regions which do not scale well for large-scale problems. As a result, the flat learning concept usually works well with large base tasks and need to be supplemented with an episodic memory to address the CF problem. The concept of wide learning [19] is integrated here to drive the flat regions to be wide, thereby increasing the chance of overlapping regions across all the tasks to be discovered. The flat local optimum regions $\Phi - b \leq \Phi \leq \Phi + b$ can be found by introducing small random noises $-b \leq \epsilon_t \leq +b$ to model's parameters multiple times such that similar but different losses are returned. Such losses can be minimized simultaneously to discover the flat minimum regions. Formally speaking, our interest is to minimize the expected loss function with respect to the joint distribution of data z and noise ϵ, $-b \leq \epsilon \leq +b$

$$\begin{aligned} R(\Phi) &= \int_{\Re^{d_\epsilon}}\int_{\Re^{d_z}} \mathcal{L}(z; \Phi + \epsilon, \Psi) dP(z) dP(\epsilon) \\ &= \mathbb{E}[\mathcal{L}(z; \Phi + \epsilon, \Psi)] \end{aligned} \tag{4}$$

where $P(z)$, $P(\epsilon)$, respectively, stand for the data distribution and the noise distribution. In practice, expected losses are impossible to be minimized, so its empirical version is

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formulated as follows:

$$\mathcal{L}_{k=1} = \frac{1}{T}\sum_{t=1}^{T}\mathcal{L}_{\text{base}}(z;\Phi+\epsilon_t,\Psi)$$
$$\mathcal{L}_{\text{base}} = \mathbb{E}_{(x)\backsim\mathcal{T}_1}\big[\mathcal{L}_{\text{psl}}(z;\Phi+\epsilon_t,\Psi,C)+\lambda_1\mathcal{L}_{\text{psa}}(z;\Phi+\epsilon_t,\Psi) +\lambda_2\mathcal{L}_{KL}(g_\Psi(f_\Phi(x)),P_U)\big]+\lambda_3\big\|C_c-C_c^*\big\|_2 \tag{5}$$

where λ_1, λ_2, and λ_3 denote predefined constants. $\mathcal{L}_{KL}(.)$ is appended to promote wide local optima where latent features are projected to P_U, the uniform distribution. That is, the uniform distribution is selected because it offers the upper bound of divergence $\log M$. C_c, C_c^* denote the latent clusters before and after noise injections where the last term is included to avoid the cluster's drifts. Note that we follow [69] where the target network $f_{\Phi'}(.)$ is momentum updated to allow stable representations. That is, the online network $f_\Phi(.)$ is updated first minimizing (5) and applied to the moving average formula to adjust the target network.

V. Few-Shot Learning Phase $k > 1$

The few-shot learning phase features data scarcity issues where it is configured under the N-way K-shot configuration. UNISA applies the feature-space data augmentation strategy to overcome this issue while putting forward the regularization-based approach to handle the CF problem.

A. Ball Data Generator

The concept of ball data generator [20] is modified to fit images rather than texts and also in the unsupervised manner. Synthetic samples of the ball data generator are induced within the smallest enclosing balls of the feature space to prevent out of distributions. Since three exist very few labels, the ball $\zeta(C_c,\sigma_c)$ is constructed via clusters in the unsupervised manner and the episodic training phase as practiced in the few-shot learning domain is absent in the FSCL problem. (C_c,σ_c) denote the centroid and standard deviation of the cth cluster estimated via the k-means clustering procedure. A synthetic sample $\tilde{z}\in\Re^{\hat{D}}$ is generated as follows:

$$\tilde{z} = C_c + u^{\frac{1}{D}}\sigma_c\frac{\omega}{||\omega||_2} \tag{6}$$

where $u\backsim\mathcal{U}(0,1)$ denotes a shifting constant, and $\omega\in\Re^{\hat{D}}\backsim\mathcal{N}(0,1)$ labels a random vector. $\tilde{z}$ is not directly injected to the training process rather fed to a projection module $\tilde{z}=\chi_W(\tilde{z})\in\Re^{\hat{D}}:\hat{z}\rightarrow\tilde{z}$ to generate stable features. The projection module is trained in such a way to avoid synthetic sample biases via the triplet loss

$$\mathcal{L}_{\text{ball}} = \sum_{\substack{\tilde{z}\notin C_j\\ \tilde{z}\in C_i}}\lambda_4\max\big\{0,d\big(\tilde{z},C_i\big)+r-d\big(\tilde{z},C_j\big)\big\} \tag{7}$$

where r, λ_4 are predefined constants, respectively. The goal is such that synthetic samples stay close to its centroid while pushing them away from other centroids where the distance margin is controlled by r. C_i, C_j are the centroid of the ith and jth clusters, respectively.

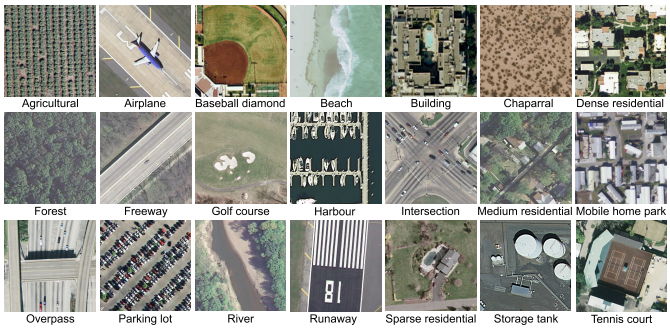


Fig. 4. Visualization of the UCMerced dataset.

B. Regularization-Based Approach

UNISA relies on the regularization-based approach to combat the CF problem due to sequences of the few-shot learning task. We apply the memory-aware synapses (MASs) [26] which can be computed in the online and unsupervised manner. Suppose that $\Theta=\{\Phi,\Psi\}$, the MAS technique is derived as follows:

$$\mathcal{R}_{\text{MAS}} = \lambda_5\sum_i\Gamma_i^{k-1}\big(\Theta_i^k-\Theta^{k-1}\big)^2 \tag{8}$$

where Γ is a parameter importance matrix, and λ_5 is a predefined constant. MAS is preferred over EWC [24] and SI [25], because it is faster to compute than them and applicable for unsupervised scenarios. We do not impose any regularization to the transformation module $\chi_W(.)$ because it only focuses on the current concept. Equation (8) can be perceived as L_2 regularization except the presence of the difference across two consecutive weight matrices Θ^{k-1}, Θ^k.

C. Flat-Wide Few-Shot Learning Phases $k > 1$

The loss function of the few-shot learning phases is written as follows:

$$\mathcal{L}_{k>1} = \mathbb{E}_{(x)\backsim\mathcal{T}_k}\big[\mathcal{L}_{\text{psl}}(z;\Phi,\Psi,C)+\lambda_1\mathcal{L}_{\text{psa}}(z;\Phi,\Psi) +\lambda_2\mathcal{L}_{KL}(g_\Psi(f_\Phi(x)),P_U)+\lambda_4\mathcal{L}_{\text{ball}}\big] +\mathcal{R}_{\text{MAS}}. \tag{9}$$

That is, the prototype scattering loss $\mathcal{L}_{\text{psl}}$ and the positive sampling alignment loss $\mathcal{L}_{\text{psa}}$ loss learn the current concept with the support of synthetic samples induced by the ball generator. The ball generator loss L_{ball} is designed to produce quality samples while L_{KL} exists to induce wide local optimum regions and drives flat regions to be wide. Finally, the regularization $\mathcal{R}_{\text{MAS}}$ addresses the CF problem. Note that the projection module is trained in the end-to-end manner alongside the online network. The flat region is retained by applying the clamping mechanism. That is, the network parameters are clamped when going beyond the flat regions. In addition, the target network is adjusted by the momentum strategy of the online network. The workflow of UNISA is pictorially depicted in Fig. 3. The pseudocode of UNISA is provided in the supplemental document.

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TABLE I

CLASSIFICATION ACCURACY ON SPLIT UCMERCED AND AID DATASETS WITH ONE- TO FIVE-SHOT SETTING AVERAGED ACROSS FIVE TIMES RUN

Method	#S	Split UCMerced						Split AID					
		Session				Avg	Gap to UNISA	Session				Avg	Gap to UNISA
		1	2	3	4			1	2	3	4		
Baseline[15]	5	18.33±0.00	17.20±0.90	14.61±1.43	11.90±0.81	15.51±0.94	-39.79±1.47	18.06±0.00	12.87±0.69	10.01±0.29	7.90±0.43	12.21±0.43	-49.21±0.71
F2M[15]		25.00±0.00	16.80±0.57	13.83±0.85	12.14±1.06	16.94±0.74	-38.36±1.35	16.49±0.00	13.38±0.25	10.10±0.17	9.67±0.37	12.41±0.24	-49.01±0.61
FSLL[12]		22.50±0.00	10.80±1.40	9.67±1.14	7.76±0.45	12.68±0.93	-42.62±1.46	25.78±0.00	12.13±2.02	8.34±0.71	7.15±0.39	13.35±1.09	-48.07±1.23
ICARL[37]		23.50±1.47	18.80±1.49	16.72±0.90	14.48±1.32	18.38±1.32	-36.92±1.74	16.41±0.31	14.58±0.73	12.47±0.64	10.15±0.71	13.40±0.62	-48.02±0.84
Rebalance[70]		12.50±0.23	10.07±0.22	8.22±0.12	7.24±0.21	9.51±0.20	-45.79±1.15	7.52±0.00	5.41±0.00	4.34±0.00	3.60±0.00	5.22±0.00	-56.20±0.56
S3C[16]		50.00±0.00	42.07±2.10	34.83±1.34	30.24±1.29	39.29±1.40	-16.01±1.80	49.27±0.00	34.08±1.09	27.94±1.03	23.92±0.47	33.80±0.79	-27.62±0.97
SCN[2]		23.75±0.00	10.47±2.01	8.83±1.56	7.76±1.02	12.70±1.37	-42.6±1.78	36.33±0.00	13.50±1.28	11.77±1.27	12.15±0.62	18.44±0.95	-42.98±1.10
ORCO[51]		42.50±0.00	13.07±2.95	15.89±2.05	15.67±2.75	21.78±2.26	-33.52±2.53	31.94±0.00	14.85±1.88	10.61±1.80	12.73±1.58	17.53±1.52	-43.89±1.62
UNISA(Ours)		**67.50±0.00**	**58.27±1.51**	**51.67±0.77**	**43.76±1.50**	**55.3±1.13**	**0.00±0.00**	**85.91±0.00**	**63.83±0.76**	**52.23±0.52**	**43.71±0.63**	**61.42±0.56**	**0.00±0.00**
Baseline[15]	4	18.33±0.00	17.33±1.12	10.44±2.83	8.86±2.22	13.74±1.88	-38.37±2.62	18.06±0.00	13.00±0.84	9.49±0.27	8.20±0.09	12.19±0.44	-49.26±0.70
F2M[15]		25.00±0.00	16.80±0.88	13.83±1.32	11.86±0.76	16.87±0.88	-35.24±2.03	16.49±0.00	13.51±0.80	9.98±0.56	9.08±0.61	12.27±0.58	-49.18±0.80
FSLL[12]		22.50±0.00	10.47±0.92	8.61±0.90	8.05±0.81	12.41±0.76	-39.7±1.98	25.78±0.00	12.21±1.73	9.58±1.86	7.85±0.61	13.86±1.31	-47.59±1.42
ICARL[37]		23.92±1.28	6.67±0.00	5.56±0.00	4.76±0.00	10.23±0.64	-41.88±1.94	16.56±0.43	5.41±0.00	4.34±0.00	3.60±0.00	7.48±0.22	-53.97±0.59
Rebalance[70]		12.75±0.44	9.73±0.39	8.39±0.18	7.10±0.20	9.49±0.32	-42.62±1.86	7.52±0.00	5.41±0.00	4.34±0.00	3.60±0.00	5.22±0.00	-56.23±0.55
S3C[16]		50.00±0.00	41.47±0.68	34.44±2.05	29.00±1.26	38.73±1.25	-13.38±2.22	49.27±0.00	33.87±0.72	27.29±1.14	23.19±0.85	33.41±0.80	-28.04±0.97
SCN[2]		23.75±0.00	10.73±1.02	9.44±1.91	7.57±1.06	12.87±1.21	-39.24±2.19	36.33±0.00	11.92±1.24	11.14±0.97	11.70±0.89	17.77±0.90	-43.68±1.05
ORCO[51]		42.50±0.00	11.40±2.32	18.78±1.38	17.67±3.44	22.59±2.19	-29.52±2.85	31.94±0.00	15.36±1.39	10.04±1.96	10.16±1.94	16.88±1.54	-44.57±1.64
UNISA(Ours)		**67.50±0.00**	**55.47±2.62**	**45.61±0.89**	**39.86±2.41**	**52.11±1.83**	**0.00±0.00**	**85.91±0.00**	**63.42±0.80**	**52.93±0.74**	**43.54±0.21**	**61.45±0.55**	**0.00±0.00**
Baseline[15]	3	18.33±0.00	17.87±2.16	10.06±2.10	9.29±1.34	13.89±1.65	-39.08±2.01	18.06±0.00	12.01±0.80	9.31±0.43	7.91±0.71	11.82±0.58	-48.71±0.69
F2M[15]		25.00±0.00	16.93±0.98	12.56±1.47	10.81±0.82	16.33±0.97	-36.64±1.5	16.49±0.00	13.17±1.13	10.17±1.22	9.28±0.72	12.28±0.91	-48.25±0.99
FSLL[12]		22.50±0.00	10.93±1.05	9.61±1.01	7.81±0.24	12.71±0.74	-40.26±1.37	25.78±0.00	13.08±1.07	9.80±1.76	9.04±1.09	14.43±1.17	-46.10±1.23
ICARL[37]		23.50±0.50	6.67±0.00	5.56±0.00	4.76±0.00	10.12±0.25	-42.85±1.18	16.62±0.37	5.41±0.00	4.34±0.00	3.60±0.00	7.49±0.19	-53.04±0.42
Rebalance[70]		12.50±0.23	9.80±0.44	8.56±0.18	7.14±0.19	9.50±0.28	-43.47±1.18	7.52±0.00	5.41±0.00	4.34±0.00	3.60±0.00	5.22±0.00	-55.31±0.38
S3C[16]		50.00±0.00	42.13±0.68	34.78±0.60	28.95±0.86	38.97±0.62	-14.00±1.31	49.27±0.00	32.79±1.20	26.45±0.86	22.56±0.59	32.77±0.79	-27.76±0.88
SCN[2]		23.75±0.00	10.80±1.01	7.89±1.34	6.19±1.79	12.16±1.23	-40.81±1.68	36.33±0.00	11.38±1.66	9.93±1.16	10.04±1.18	16.92±1.17	-43.61±1.23
ORCO[51]		42.50±0.00	12.87±0.88	17.44±2.64	15.57±2.83	22.1±1.98	-30.87±2.29	31.94±0.00	12.22±1.62	9.41±1.40	10.62±0.69	16.05±1.12	-44.48±1.18
UNISA(Ours)		**67.50±0.00**	**56.47±1.85**	**48.28±1.16**	**39.62±0.72**	**52.97±1.15**	**0.00±0.00**	**85.91±0.00**	**62.46±0.34**	**51.06±0.28**	**42.68±0.61**	**60.53±0.38**	**0.00±0.00**
Baseline[15]	2	18.33±0.00	15.00±2.76	9.67±0.98	7.57±0.99	12.64±1.55	-39.42±1.88	18.06±0.00	10.75±0.85	9.30±1.12	8.12±0.50	11.56±0.75	-49.28±0.87
F2M[15]		25.00±0.00	17.33±0.69	11.94±0.22	10.71±0.71	16.25±0.51	-35.81±1.19	16.49±0.00	13.33±0.79	10.33±0.38	9.04±0.41	12.3±0.48	-48.54±0.66
FSLL[12]		22.50±0.00	9.87±0.54	9.56±0.73	7.90±0.82	12.46±0.61	-39.6±1.23	25.78±0.00	11.83±0.90	9.78±0.88	7.69±1.70	13.77±1.06	-47.07±1.15
ICARL[37]		25.42±1.41	6.67±0.00	5.56±0.00	4.76±0.00	10.60±0.71	-41.46±1.28	16.58±0.46	5.41±0.00	4.34±0.00	3.60±0.00	7.48±0.23	-53.36±0.51
Rebalance[70]		12.50±0.57	10.13±0.23	8.50±0.19	7.14±0.13	9.57±0.33	-42.49±1.12	7.52±0.00	5.41±0.00	4.34±0.00	3.60±0.00	5.22±0.00	-55.62±0.45
S3C[16]		50.00±0.00	40.60±1.19	35.22±1.00	28.62±1.65	38.61±1.13	-13.45±1.56	49.27±0.00	33.69±0.30	26.20±0.98	22.81±0.65	32.99±0.61	-27.85±0.76
SCN[2]		23.75±0.00	8.20±1.45	7.39±1.68	6.67±1.02	11.50±1.22	-40.56±1.62	36.33±0.00	9.49±1.41	7.10±0.92	8.79±1.10	15.43±1.01	-45.41±1.11
ORCO[51]		42.50±0.00	11.07±0.91	16.22±1.51	15.81±2.19	21.40±1.41	-30.66±1.77	31.94±0.00	11.31±2.32	8.61±1.50	10.52±3.37	15.6±2.18	-45.24±2.23
UNISA(Ours)		**67.50±0.00**	**55.00±1.21**	**46.39±1.38**	**39.33±1.11**	**52.06±1.07**	**0.00±0.00**	**85.91±0.00**	**63.20±0.39**	**51.05±0.69**	**43.20±0.41**	**60.84±0.45**	**0.00±0.00**
Baseline[15]	1	18.33±0.00	12.33±2.89	9.67±0.93	8.19±0.67	12.13±1.55	-38.39±2.32	18.06±0.00	11.53±0.80	9.59±0.35	8.13±0.15	11.83±0.44	-48.10±0.79
F2M[15]		25.00±0.00	18.07±3.34	12.28±1.04	10.90±1.21	16.56±1.85	-33.96±2.53	16.49±0.00	13.68±0.94	9.95±0.45	8.88±0.48	12.25±0.57	-47.68±0.87
FSLL[12]		22.50±0.00	10.40±1.02	8.06±0.74	7.05±0.74	12.00±0.73	-38.52±1.87	25.78±0.00	11.04±1.00	9.84±0.68	8.90±1.19	13.89±0.85	-46.04±1.08
ICARL[37]		23.92±0.85	6.67±0.00	5.56±0.00	4.76±0.00	10.23±0.43	-40.29±1.77	16.47±0.33	5.41±0.00	4.34±0.00	3.60±0.00	7.46±0.17	-52.47±0.68
Rebalance[70]		12.58±0.15	9.87±0.30	8.50±0.12	7.29±0.21	9.56±0.21	-40.96±1.73	7.52±0.00	5.41±0.00	4.34±0.00	3.60±0.00	5.22±0.00	-54.71±0.66
S3C[16]		50.00±0.00	40.33±1.00	34.89±1.26	27.67±1.24	38.22±1.02	-12.30±2.00	49.27±0.00	32.00±1.26	24.80±0.29	20.56±0.70	31.66±0.74	-28.27±0.99
SCN[2]		23.75±0.00	7.40±1.58	6.61±1.70	5.76±0.79	10.88±1.23	-39.64±2.11	36.33±0.00	8.48±1.02	6.43±0.51	7.41±0.61	14.66±0.65	-45.27±0.93
ORCO[51]		42.50±0.00	12.87±1.04	14.61±1.91	8.86±1.63	19.71±1.36	-30.81±2.19	31.94±0.00	13.50±1.16	8.53±0.64	7.58±0.82	15.39±0.78	-44.14±1.02
UNISA(Ours)		**67.50±0.00**	**55.33±0.58**	**43.00±2.56**	**36.24±2.22**	**50.52±1.72**	**0.00±0.00**	**85.91±0.00**	**62.31±0.47**	**50.19±0.32**	**41.31±1.20**	**59.93±0.66**	**0.00±0.00**

VI. EXPERIMENTS

This section presents numerical validations of UNISA, including experiments on RS problems, comparisons against prior arts, ablation studies to demonstrate the advantage of each learning module, and sensitivity analysis. Source codes of UNISA are shared publicly in https://github.com/anwarmaxsum/UNISA.

A. Experimental Setting

All the consolidated algorithms are benchmarked in six RS datasets: the UC-Merced [71], the AID [72], the RSICB-256 [73], the RESISC45 [74], the EuroSAT [75], and the HyRank [7] under one-, two-, three-, four-, and five-shot configurations.

1) Dataset: The UC Merced dataset comprises 21 classes where each class carries 100 images. It is split into four tasks where the base task possesses 12 classes while each few-shot task includes three classes. The spatial resolution of this dataset is 0.3 m. Visualization of the UC Merced problem is offered in Fig. 4. The AID dataset presents 30 categories where each category has around 300 images with the spatial resolution between 0.5 and 8 m. This dataset is divided into four tasks where base task encompasses 15 classes while each few-shot task consists of five classes. The dataset, RSICB256, has 35 classes where each class comprises 800 images with the spatial resolution between 0.22 and 3 m. This dataset is set to four tasks where the base task consists of 20 classes and each few-shot task carries five classes. The RESISC45 dataset carries 45 classes where each class contains 700 images with the spatial resolution between 0.2 m and more than 30 m/px. This dataset is set to four tasks where the base task consists of 24 classes and each few-shot task carries seven classes. The EuroSAT dataset portrays the land-use classification problem of the Sentinel 2 satellite. This dataset comprises ten classes where the base task covers four classes while each few-shot task carries two classes. The HyRank dataset is a hyperspectral dataset developed by the International Society for Photogrammetry and Remote Sensing scientific initiatives [7]. The dataset is collected by the Hyperion sensor having 176 spectral bands. It consists of two scenes, Dioni and Loukia, where we focus only on the Loukia scene here having 249×945 pixels. This dataset comprises 12 classes where the first six classes are reserved for the base class while the few-shot tasks carry

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TABLE II
CLASSIFICATION ACCURACY ON THE SPLIT RSICB256 AND SPLIT RESISC45 DATASETS WITH ONE- TO FIVE-SHOT SETTING AVERAGED ACROSS FIVE TIMES RUN

Method	#S	Split RSICB256						Split RESISC45					
		Session				Avg	Gap to UNISA	Session				Avg	Gap to UNISA
		1	2	3	4			1	2	3	4		
Baseline[15]	5	41.71±0.00	33.18±0.47	28.39±1.07	28.50±0.61	32.95±0.66	-44.12±0.98	16.93±0.00	11.24±0.00	10.21±0.00	8.63±0.00	11.75±0.00	-54.66±0.00
F2M[15]		21.78±0.00	18.19±0.73	16.25±0.63	14.64±1.16	17.72±0.75	-59.35±1.04	16.52±0.00	13.00±0.00	10.92±0.00	9.33±0.00	12.44±0.00	-53.97±0.00
FSLL[12]		42.21±0.00	24.18±2.05	21.02±0.55	23.28±0.90	27.67±1.15	-49.40±1.36	16.70±0.00	11.43±0.00	8.46±0.00	7.05±0.00	10.91±0.00	-55.50±0.00
ICARL[37]		31.91±0.62	2.23±0.00	1.76±0.00	1.43±0.00	9.33±0.31	-67.74±0.78	16.92±0.18	3.23±0.00	2.63±0.00	2.22±0.00	6.25±0.09	-60.16±0.09
Rebalance[70]		6.54±0.06	5.33±0.12	4.25±0.03	3.45±0.03	4.89±0.07	-72.18±0.72	4.32±0.02	3.31±0.01	2.70±0.01	2.28±0.01	3.15±0.01	-63.26±0.01
S3C[16]		82.74±0.00	69.33±0.41	62.10±0.71	57.23±0.21	67.85±0.42	-9.22±0.83	47.11±0.00	36.71±0.64	30.41±0.21	26.04±0.24	35.07±0.36	-31.34±0.36
SCN[2]		33.60±0.00	8.32±0.42	17.42±0.59	14.18±1.30	18.38±0.74	-58.69±1.03	14.61±0.00	5.37±0.46	6.48±0.38	6.63±0.56	8.27±0.41	-58.14±0.41
ORCO[51]		40.83±0.00	31.67±0.00	15.66±0.00	21.12±0.00	27.10±0.00	-49.97±1.16	34.35±0.00	8.86±0.94	10.67±1.09	10.06±1.12	15.99±0.91	-50.43±0.91
UNISA(Ours)		**95.41±0.00**	**80.36±0.83**	**70.08±0.69**	**62.43±0.94**	**77.07±0.72**	**0.00±0.00**	**88.51±0.00**	**69.56±0.21**	**57.93±0.18**	**49.62±0.71**	**66.41±0.38**	**0.00±0.00**
Baseline[15]	4	41.71±0.00	33.00±0.12	28.41±0.75	28.13±0.68	32.81±0.51	-43.41±1.05	16.93±0.00	11.89±0.00	10.98±0.00	9.10±0.00	12.23±0.00	-53.83±0.00
F2M[15]		21.78±0.00	18.19±0.65	16.52±0.36	15.32±1.16	17.95±0.69	-58.27±1.15	16.01±0.00	13.29±0.00	11.45±0.00	9.08±0.00	12.46±0.00	-53.59±0.00
FSLL[12]		42.21±0.00	24.18±2.05	21.02±0.55	23.28±0.90	27.67±1.15	-48.55±1.47	16.55±0.00	11.29±0.00	7.80±0.00	6.63±0.00	10.57±0.00	-55.48±0.00
ICARL[37]		32.42±0.49	2.23±0.00	1.76±0.00	1.43±0.00	9.46±0.25	-66.76±0.95	17.05±0.17	3.23±0.00	2.63±0.00	2.22±0.00	6.28±0.09	-59.77±0.09
Rebalance[70]		6.54±0.07	5.42±0.08	4.20±0.07	3.43±0.04	4.90±0.07	-71.32±0.92	4.29±0.02	3.32±0.01	2.71±0.00	2.29±0.00	3.15±0.01	-62.90±0.01
S3C[16]		82.74±0.00	68.81±0.32	61.30±0.77	56.00±0.42	67.21±0.47	-9.01±1.03	47.11±0.00	36.82±0.47	30.50±0.54	25.90±0.52	35.08±0.44	-30.97±0.44
SCN[2]		33.60±0.00	8.74±0.66	16.33±1.08	13.31±1.03	18.00±0.82	-58.22±1.23	14.61±0.00	5.29±0.43	6.64±0.62	6.04±0.48	8.15±0.45	-57.91±0.45
ORCO[51]		40.83±0.00	31.57±0.00	26.42±0.00	25.45±0.00	28.21±0.00	-48.01±1.38	34.35±0.00	10.12±0.76	8.89±1.82	9.38±0.57	15.69±1.03	-50.37±1.03
UNISA(Ours)		**95.41±0.00**	**80.15±0.60**	**68.69±1.08**	**60.63±1.36**	**76.22±0.92**	**0.00±0.00**	**88.51±0.00**	**69.36±0.40**	**57.42±0.46**	**48.90±0.45**	**66.05±0.38**	**0.00±0.00**
Baseline[15]	3	41.71±0.00	32.78±0.33	27.89±1.26	27.35±1.46	32.43±0.98	-42.34±1.23	17.05±0.00	11.96±0.00	10.60±0.00	8.86±0.00	12.12±0.00	-53.29±0.00
F2M[15]		21.78±0.00	18.26±0.94	16.68±0.75	14.84±1.23	17.89±0.86	-56.88±1.13	16.07±0.00	12.10±0.00	11.13±0.00	9.79±0.00	12.27±0.00	-53.14±0.00
FSLL[12]		42.21±0.00	22.56±2.15	19.92±1.13	22.14±1.30	26.71±1.38	-48.06±1.57	16.34±0.00	10.51±0.00	8.89±0.00	7.44±0.00	10.80±0.00	-54.62±0.00
ICARL[37]		32.52±0.47	2.23±0.00	1.76±0.00	1.43±0.00	9.49±0.24	-65.28±0.78	16.98±0.09	3.23±0.00	2.63±0.00	2.22±0.00	6.27±0.05	-59.15±0.05
Rebalance[70]		6.58±0.13	5.35±0.04	4.22±0.05	3.51±0.03	4.92±0.07	-69.85±0.74	4.29±0.02	3.33±0.01	2.70±0.01	2.29±0.01	3.15±0.01	-62.26±0.01
S3C[16]		82.74±0.00	69.35±0.49	61.02±0.90	55.13±0.72	67.06±0.63	-7.71±0.97	47.11±0.00	36.13±0.62	29.92±0.55	25.06±0.40	34.56±0.46	-30.86±0.46
SCN[2]		33.60±0.00	8.65±0.30	15.10±1.18	12.09±0.87	17.36±0.75	-57.41±1.05	14.61±0.00	5.40±0.13	5.89±0.52	6.02±0.94	7.98±0.54	-57.43±0.54
ORCO[51]		40.83±0.00	27.70±0.00	25.21±0.00	27.39±0.00	27.61±0.00	-47.16±1.35	34.35±0.00	8.69±1.02	9.00±1.19	7.99±1.62	15.01±1.13	-50.40±1.13
UNISA(Ours)		**95.41±0.00**	**78.77±0.79**	**66.31±0.61**	**58.58±1.09**	**74.77±0.74**	**0.00±0.00**	**88.51±0.00**	**68.82±0.56**	**56.29±0.81**	**48.03±0.74**	**65.41±0.62**	**0.00±0.00**
Baselin[15]	2	41.71±0.00	32.49±0.86	26.67±0.68	25.69±1.36	31.64±0.87	-42.43±1.26	16.73±0.00	11.24±0.00	9.94±0.00	7.98±0.00	11.47±0.00	-53.95±0.00
F2M[15]		21.78±0.00	17.65±0.66	15.82±1.05	14.85±1.62	17.53±1.02	-56.54±1.37	15.48±0.00	11.77±0.00	10.28±0.00	8.94±0.00	11.62±0.00	-53.80±0.00
FSLL[12]		42.21±0.00	22.03±1.86	19.23±2.01	21.73±1.14	26.3±1.48	-47.77±1.74	16.93±0.00	9.65±0.00	8.14±0.00	6.13±0.00	10.21±0.00	-55.21±0.00
ICARL[37]		32.28±0.60	2.23±0.00	1.76±0.00	1.43±0.00	9.43±0.30	-64.64±0.96	17.10±0.12	3.23±0.00	2.63±0.00	2.22±0.00	6.30±0.06	-59.13±0.06
Rebalance[70]		6.57±0.11	5.40±0.06	4.20±0.08	3.44±0.05	4.90±0.08	-69.17±0.91	4.30±0.01	3.32±0.01	2.71±0.01	2.29±0.01	3.16±0.01	-62.27±0.01
S3C[16]		82.74±0.00	68.73±0.71	58.97±1.43	53.32±1.02	65.94±0.95	-8.13±1.32	47.11±0.00	36.09±0.58	29.11±0.16	24.63±0.47	34.24±0.38	-31.19±0.38
SCN[2]		33.60±0.00	7.15±1.61	13.62±1.20	10.77±1.22	16.29±1.17	-57.78±1.48	14.61±0.00	5.25±0.55	4.83±0.63	5.03±0.57	7.43±0.51	-57.99±0.51
ORCO[51]		40.83±0.00	24.84±0.00	23.58±0.00	5.38±0.00	25.99±0.,00	-48.08±1.83	34.35±0.00	8.58±1.27	7.88±2.35	8.20±1.74	14.75±1.59	-50.67±1.59
UNISA(Ours)		**95.41±0.00**	**79.19±1.00**	**65.19±0.91**	**56.49±1.21**	**74.07±0.91**	**0.00±0.00**	**88.51±0.00**	**68.53±0.66**	**56.65±0.39**	**47.99±0.65**	**65.42±0.50**	**0.00±0.00**
Baseline[15]	1	41.71±0.00	32.50±0.62	26.13±1.18	24.40±0.73	31.19±0.76	-40.68±1.21	16.61±0.00	12.47±0.00	10.58±0.00	7.94±0.00	11.90±0.00	-52.99±0.00
F2M[15]		21.78±0.00	16.33±1.44	15.54±0.79	13.51±0.74	16.79±0.90	-55.08±1.30	16.76±0.00	11.82±0.00	9.42±0.00	8.16±0.00	11.54±0.00	-53.35±0.00
FSLL[12]		42.21±0.00	19.69±1.78	18.07±1.96	19.78±2.47	24.94±1.81	-46.93±2.04	16.37±0.00	8.80±0.00	7.78±0.00	4.86±0.00	9.45±0.00	-55.44±0.00
ICARL[37]		32.76±0.36	2.23±0.00	1.76±0.00	1.43±0.00	9.55±0.18	-62.32±0.96	17.17±0.28	3.23±0.00	2.63±0.00	2.22±0.00	6.31±0.14	-58.58±0.14
Rebalance[70]		6.56±0.08	5.38±0.04	4.21±0.05	3.47±0.05	4.91±0.06	-66.96±0.94	4.30±0.01	3.33±0.02	2.72±0.01	2.29±0.01	3.16±0.01	-61.73±0.01
S3C[16]		82.74±0.00	67.37±0.77	58.27±0.99	51.83±0.71	65.05±0.72	-6.82±1.18	47.11±0.00	35.65±1.07	27.97±0.75	22.93±0.82	33.42±0.77	-31.48±0.77
SCN[2]		33.60±0.00	5.43±0.80	8.56±2.04	8.31±1.83	13.98±1.43	-57.89±1.71	14.61±0.00	4.63±0.45	4.17±0.98	3.85±0.42	6.82±0.58	-58.08±0.58
ORCO[51]		40.83±0.00	10.72±0.00	13.61±0.00	14.79±0.00	20.55±0.00	-51.32±1.22	34.35±0.00	7.81±1.29	5.97±0.76	5.95±0.34	13.52±0.77	-51.37±0.77
UNISA(Ours)		**95.41±0.00**	**77.68±0.71**	**62.37±1.12**	**52.02±1.32**	**71.87±0.94**	**0.00±0.00**	**88.51±0.00**	**68.26±0.47**	**55.68±0.57**	**47.09±0.57**	**64.89±0.47**	**0.00±0.00**

two classes totaling three few-shot tasks. We only apply a maximum 10% (25 samples) label proportion of the base task while the few-shot tasks exploit the full label proportions due to the data scarcity problems. One to five shots cases are attempted in our experiments. Note that labeled samples are only used for inferences in UNISA while the training process itself is free of labeled samples.

2) Baseline: UNISA is compared against iCaRL [37], Rebalance [70], FSLL [12], F2M [15], S3C [16], and ORCO [51] representing prior arts of the FSCL algorithms. In addition, baseline is added and reflects the case where a network is frozen while leaving only the prototype construction in the few-shot learning phase. We also compare our approach with the state-of-the-art continual RS learning, namely, asymmetric collaborative network (SCN) [2]. Since these algorithms are fully supervised in nature, they are only trained with 10% labeled samples of the base task. There are no existing algorithms designed specifically for unsupervised FSCL problems. All the consolidated algorithms are executed under the same computational environments, single NVDIA A100 GPU with 40-GB RAM, using their official implementations. Note that the competitor methods are trained in a supervised way using all the available labels both in the base task and the few-shot task, while our proposed method (UNISA) is trained in an unsupervised way using unlabeled samples in both the tasks. UNISA only uses labels for inference (testing).

3) Hyperparameters: we follow the same hyperparameters' setting from F2M [15] for competitor methods and UNISA. UNISA also follows the same hyperparameters as [18].

a) Base task: F2M is run with 1000 epochs, 0.1 initial learning rate, 0.01 bound value, and two random times. Baseline and FSLL are run with 240 epochs and 0.1 initial learning rate. ICARL and Rebalance are run with 130 epochs and 0.1 initial learning rate. UNISA is run with 1000 epochs, 0.05 initial learning rate, 0.01 bound value, and two random times.

b) Few-shot task: F2M is run with six epochs and 0.02 initial learning rate. The baseline model only generates prototypes (not updated) in the few-shot task. FSLL is run with 16 epochs and 0.001 initial learning rate. ICARL and Rebalance are run with 90 epochs and 0.01 initial learning

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TABLE III

CLASSIFICATION ACCURACY ON THE SPLIT EUROSAT AND HYRANK DATASETS WITH ONE- TO FIVE-SHOT SETTING AVERAGED ACROSS FIVE TIMES RUN

Method	#S	Split EuroSAT						Split HyRank					
		Session				Avg	Gap to UNISA	Session				Avg	Gap to UNISA
		1	2	3	4			1	2	3	4		
Baseline[15]	5	53.39±0.00	46.77±0.85	34.66±0.86	27.61±0.97	40.61±0.78	-33.04±1.95	33.42±0.00	26.56±3.59	21.51±2.33	23.37±2.03	26.22±2.37	-17.48±2.65
F2M[15]		56.22±0.00	43.12±1.10	33.91±1.61	27.53±1.52	40.20±1.24	-33.45±2.18	16.79±0.00	10.27±2.44	9.26±2.36	5.43±1.15	10.44±1.79	-33.26±2.14
FSLL[12]		57.78±0.00	30.73±2.26	21.82±2.22	23.14±2.44	33.37±2.00	-40.28±2.68	35.27±0.00	26.12±1.46	21.54±2.10	23.91±0.53	26.71±1.31	-16.99±1.76
ICARL[37]		36.48±0.15	30.58±1.99	23.70±2.61	20.54±1.72	27.83±1.85	-45.82±2.57	26.78±1.03	31.29±0.00	29.44±0.00	29.06±0.00	29.14±0.52	-14.56±1.29
Rebalance[70]		26.09±0.00	18.75±0.00	13.95±0.00	11.11±0.00	17.48±0.00	-56.17±1.79	34.31±0.00	31.29±0.00	29.44±0.00	29.06±0.00	31.03±0.00	-12.67±1.18
S3C[16]		84.26±0.00	65.87±0.28	52.31±1.62	43.53±1.16	61.49±1.01	-12.16±2.06	49.86±0.00	34.31±0.83	30.71±0.60	28.76±1.03	35.91±0.73	-7.79±1.39
SCN[2]		67.35±0.00	27.52±1.07	18.21±3.61	16.66±2.15	32.44±2.17	-41.21±2.81	47.48±0.00	8.74±0.06	8.16±1.11	5.10±1.56	17.37±0.96	-26.33±1.52
ORCO[51]		71.70±0.00	27.28±0.23	20.66±3.41	20.57±3.53	35.05±2.46	-38.60±3.04	36.37±0.00	7.86±0.59	4.65±1.36	1.70±0.84	12.65±0.85	-31.05±1.45
UNISA(Ours)		**96.70±0.00**	**79.06±1.71**	**64.95±2.45**	**53.89±1.98**	**73.65±1.79**	**0.00±0.00**	**52.75±0.00**	**44.56±0.51**	**39.84±1.39**	**37.64±1.83**	**43.70±1.18**	**0.00±0.00**
Baseline[15]	4	53.39±0.00	46.99±0.45	34.20±0.67	27.46±0.64	40.51±0.52	-31.76±1.74	33.42±0.00	27.62±2.36	23.13±1.91	19.72±3.02	25.97±2.14	-17.75±2.48
F2M[15]		56.22±0.00	42.66±1.15	32.45±1.09	27.06±0.80	39.60±0.89	-32.67±1.88	16.79±0.00	11.05±1.12	7.30±1.90	4.43±1.51	9.89±1.34	-33.83±1.84
FSLL[12]		57.78±0.00	39.43±0.65	26.43±2.44	26.16±1.39	37.45±1.44	-34.82±2.20	35.27±0.00	25.39±0.60	23.62±1.52	22.65±1.77	26.73±1.20	-16.99±1.74
ICARL[37]		36.54±0.12	18.75±0.00	13.95±0.00	11.11±0.00	20.09±0.06	-52.18±1.66	25.77±0.67	31.29±0.00	29.44±0.00	29.06±0.00	28.89±0.34	-14.83±1.31
Rebalance[70]		26.09±0.00	18.75±0.00	13.95±0.00	11.11±0.00	17.48±0.00	-54.79±1.66	34.31±0.00	31.29±0.00	29.44±0.00	29.06±0.00	31.03±0.00	-12.69±1.26
S3C[16]		84.26±0.00	66.11±0.93	53.20±1.40	44.13±0.89	61.93±0.95	-10.34±1.91	49.86±0.00	34.49±0.86	29.74±1.20	28.62±1.14	35.68±0.93	-8.04±1.57
SCN[2]		67.35±0.00	27.20±0.72	18.40±2.98	13.10±1.67	31.51±1.75	-40.76±2.41	47.39±0.00	8.79±0.00	6.62±1.46	3.13±0.80	16.48±0.83	-27.24±1.51
ORCO[51]		71.70±0.00	26.41±1.73	26.27±4.56	19.08±1.61	35.87±2.57	-36.40±3.06	36.37±0.00	7.86±0.59	4.65±1.36	1.70±0.84	12.65±0.85	-31.07±1.50
UNISA(Ours)		**96.70±0.00**	**78.42±0.95**	**62.93±2.06**	**51.01±2.43**	**72.27±1.66**	**0.00±0.00**	**52.75±0.00**	**45.15±0.97**	**39.85±1.26**	**37.14±1.96**	**43.72±1.26**	**0.00±0.00**
Baseline[15]	3	53.39±0.00	46.48±0.98	33.54±1.60	27.80±0.73	40.30±1.01	-30.92±2.69	33.42±0.00	29.85±0.58	23.95±2.22	23.01±2.67	27.56±1.76	-16.46±1.91
F2M[15]		56.22±0.00	42.56±1.07	32.61±0.67	26.51±0.63	39.48±0.71	-31.74±2.59	16.79±0.00	9.57±2.46	4.95±1.28	5.32±1.24	9.16±1.52	-34.86±1.69
FSLL[12]		57.78±0.00	39.48±1.37	27.23±0.94	25.77±1.14	37.57±1.01	-33.65±2.69	35.27±0.00	26.05±1.71	23.88±0.37	23.81±0.82	27.25±0.97	-16.77±1.22
ICARL[37]		36.63±0.20	18.75±0.00	13.95±0.00	11.11±0.00	20.11±0.10	-51.11±2.49	26.98±1.73	31.29±0.00	29.44±0.00	29.06±0.00	29.19±0.87	-14.83±1.14
Rebalance[70]		26.09±0.00	18.75±0.00	13.95±0.00	11.11±0.00	17.48±0.00	-53.74±2.49	34.31±0.00	31.29±0.00	29.44±0.00	29.06±0.00	31.03±0.00	-12.99±0.74
S3C[16]		84.26±0.00	66.03±0.40	52.15±1.30	43.81±1.51	61.56±1.02	-9.66±2.69	49.86±0.00	35.05±0.64	30.50±0.94	28.23±1.83	35.91±1.08	-8.11±1.31
SCN[2]		67.35±0.00	26.39±0.92	19.57±2.17	17.13±1.87	32.61±1.50	-38.61±2.91	47.39±0.00	8.75±0.02	7.21±1.30	4.62±1.05	16.99±0.84	-27.03±1.12
ORCO[51]		71.70±0.00	24.26±2.64	23.63±5.92	19.44±0.41	34.76±3.25	-36.46±4.09	36.37±0.00	7.86±0.59	4.65±1.36	1.70±0.84	12.65±0.85	-31.37±1.13
UNISA(Ours)		**96.70±0.00**	**78.88±1.01**	**60.47±2.44**	**48.81±4.22**	**71.22±2.49**	**0.00±0.00**	**52.75±0.00**	**44.90±0.65**	**40.49±1.08**	**37.93±0.76**	**44.02±0.74**	**0.00±0.00**
Baseline[15]	2	53.39±0.00	45.09±0.96	32.75±2.52	26.40±1.54	39.41±1.55	-31.88±2.43	33.42±0.00	30.40±0.42	24.87±2.27	23.43±1.55	28.03±1.39	-15.61±1.55
F2M[15]		56.22±0.00	40.83±1.39	32.21±0.74	27.11±1.04	39.09±0.94	-32.20±2.09	16.79±0.00	9.87±1.83	6.46±1.15	5.53±1.19	9.66±1.23	-33.98±1.41
FSLL[12]		57.78±0.00	39.88±1.22	26.98±1.01	24.21±1.29	37.21±1.02	-34.08±2.13	35.27±0.00	25.87±2.71	24.89±1.04	23.10±1.31	27.28±1.59	-16.36±1.73
ICARL[37]		36.61±0.13	18.75±0.00	13.95±0.00	11.11±0.00	20.11±0.07	-51.18±1.87	24.09±5.44	31.29±0.00	29.44±0.00	29.06±0.00	28.47±2.72	-15.17±2.80
Rebalance[70]		26.09±0.00	18.75±0.00	13.95±0.00	11.11±0.00	17.48±0.00	-53.81±1.87	34.31±0.00	31.29±0.00	29.44±0.00	29.06±0.00	31.03±0.00	-12.61±0.68
S3C[16]		84.26±0.00	65.30±1.17	51.85±2.18	42.75±2.46	61.04±1.74	-10.25±2.55	49.86±0.00	34.54±0.99	29.61±1.20	27.58±1.15	35.4±0.97	-8.24±1.18
SCN[2]		67.35±0.00	24.76±2.23	13.67±1.36	16.29±2.64	30.52±1.86	-40.77±2.64	47.48±0.00	8.54±0.37	5.26±2.51	3.07±1.42	16.09±1.45	-27.55±1.60
ORCO[51]		71.70±0.00	22.42±2.46	20.08±2.32	19.20±5.55	33.35±3.25	-37.94±3.75	36.37±0.00	7.72±0.45	4.55±1.56	1.79±1.16	12.61±1.00	-31.03±1.21
UNISA(Ours)		**96.70±0.00**	**76.84±0.91**	**61.54±1.73**	**50.08±3.19**	**71.29±1.87**	**0.00±0.00**	**52.75±0.00**	**44.32±1.05**	**39.78±0.77**	**37.72±0.36**	**43.64±0.68**	**0.00±0.00**
Baseline[15]	1	53.39±0.00	41.69±3.05	31.10±1.80	25.56±1.34	37.94±1.89	-31.92±2.52	33.42±0.00	30.45±0.39	27.97±0.29	27.46±0.53	29.83±0.36	-15.03±1.03
F2M[15]		56.22±0.00	40.55±1.66	31.21±1.02	24.71±1.50	38.17±1.23	-31.69±2.07	16.79±0.00	10.98±1.65	8.35±1.70	6.30±0.98	10.61±1.28	-34.25±1.60
FSLL[12]		57.78±0.00	38.79±1.68	26.51±1.91	22.16±1.71	36.31±1.53	-33.55±2.26	35.27±0.00	24.75±1.99	25.20±1.56	23.97±1.01	27.30±1.36	-17.56±1.66
ICARL[37]		36.49±0.14	18.75±0.00	13.95±0.00	11.11±0.00	20.08±0.07	-49.78±1.66	21.72±7.02	31.29±0.00	29.44±0.00	29.06±0.00	27.88±3.51	-16.98±3.64
Rebalance[70]		26.09±0.00	18.75±0.00	13.95±0.00	11.11±0.00	17.48±0.00	-52.38±1.66	34.31±0.00	31.29±0.00	29.44±0.00	29.06±0.00	31.03±0.00	-13.83±0.96
S3C[16]		84.26±0.00	63.38±2.29	49.35±2.04	39.71±1.96	59.18±1.82	-10.68±2.46	49.86±0.00	33.57±3.28	27.51±2.31	26.41±1.77	34.34±2.19	-10.52±2.39
SCN[2]		67.35±0.00	24.39±1.15	14.48±1.21	10.61±1.98	29.21±1.29	-40.65±2.10	47.48±0.00	7.99±1.24	3.51±1.14	2.60±1.26	15.4±1.05	-29.46±1.42
ORCO[51]		71.70±0.00	24.67±0.99	19.03±1.25	18.32±0.77	33.43±0.89	-36.43±1.88	36.37±0.00	7.36±0.61	3.60±0.55	0.93±0.08	12.07±0.41	-32.79±1.04
UNISA(Ours)		**96.70±0.00**	**75.06±1.56**	**59.69±1.50**	**48.00±2.52**	**69.86±1.66**	**0.00±0.00**	**52.75±0.00**	**44.80±0.70**	**41.49±1.15**	**40.38±1.37**	**44.86±0.96**	**0.00±0.00**

TABLE IV

ABLATION STUDY OF THE PROPOSED METHOD WITH VARIOUS CONFIGURATIONS ON FIVE-SHOT UCMERCED DATASET, FM AND WM STAND FOR FLAT MINIMA AND WIDE MINIMA, RESPECTIVELY

Code	Configuration					Session				Avg
	FM	WM ($\mathcal{L}_{cpr}$)	$\mathcal{L}_{psl}$	$\mathcal{L}_{psa}$	$\mathcal{L}_{ball}$	1	2	3	4	
A		✓	✓	✓	✓	65.00±0.00	55.80±1.35	48.83±1.19	40.95±1.05	52.65±1.04
B	✓		✓	✓	✓	62.92±0.00	56.27±1.98	50.22±1.48	43.33±0.66	53.19±1.28
C	✓	✓		✓	✓	65.83±0.00	55.20±0.86	46.17±0.97	38.43±1.01	51.41±0.82
D	✓	✓	✓		✓	51.67±0.00	45.80±1.74	41.33±1.56	35.10±2.62	43.48±1.76
E	✓	✓	✓	✓		66.67±0.00	56.93±0.34	47.00±1.89	40.05±1.45	52.66±1.20
UNISA	✓	✓	✓	✓	✓	**67.50±0.00**	**58.27±1.51**	**51.67±0.77**	**43.76±1.50**	**55.30±1.13**

rate. UNISA is run with maximum five epochs and predicted learning rate by cosine annealing. All the methods use 0.9 momentum, $5e^{-4}$ weight decay, 128 batch size, SGD optimizer, learning rate scheduler, and ResNet18 as the base network. These hyperparameters are found using the grid search to ensure fair comparisons. λ_i for $i \in [1, 5]$ is set to 1.0 except λ_4 that is set to 0.1. The values for the coefficient are chosen so that each term in (8) is in the range [0.0, 1.0].

B. Numerical Results

Table I reports consolidated numerical results of the UCMerced problem. UNISA outperforms its counterparts with large margins in one- to five-shot settings, i.e., over 30%. This applies for both per-task accuracy and average accuracy. The same pattern is observed in the AID dataset shown in Table I. UNISA beats other algorithms with over 50% gaps for all the cases. This finding is consistent also for the RSICB256 problem, RESISC45, and the EuroSAT problem as reported in Tables II and III where UNISA delivers the most encouraging performances with large gaps, over 30% differences. In addition, UNISA performs well in the hyperspectral dataset, the HyRank dataset, with over 8% margins compared with its competitors as reported in Table III. Large margins confirm the challenging characteristics of the unsupervised FSCL which

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TABLE V

EXTENDED ABLATION STUDY OF THE PROPOSED METHOD WITH VARIOUS LOSS COEFFICIENTS I.E., $\lambda_i, i \in [1–5]$ ON FIVE-SHOT UCMERCED DATASET, * DENOTES THE RESULT OF OUR DEFAULT CONFIGURATION

Code	λ_1	λ_2	λ_3	λ_4	λ_5	Avg. Accuracy
U1	**1.0**	**1.0**	**1.0**	**0.1**	**1.0**	***55.30+-1.13**
U2	1.0	1.0	0.1	0.1	1.0	54.35+-1.47
U3	1.0	0.1	1.0	0.1	1.0	54.35+-1.47
U4	1.0	0.1	0.1	0.1	1.0	54.35+-1.47
U5	0.1	1.0	1.0	0.1	1.0	49.16+-1.02
U6	0.1	1.0	0.1	0.1	1.0	49.16+-1.02
U7	0.1	0.1	1.0	0.1	1.0	49.16+-1.02
U8	0.1	0.1	0.1	0.1	1.0	49.16+-1.02
U9	1.0	1.0	1.0	0.1	0.1	55.31+-1.14
U10	1.0	1.0	1.0	0.01	1.0	55.31+-1.14
U11	1.0	1.0	1.0	0.01	0.1	55.31+-1.14

TABLE VI

ROBUSTNESS ANALYSIS OF THE PROPOSED METHOD AGAINST VARIOUS NOISES ON THE FIVE-SHOT UCMERCED DATASET

Method	Noise	Session				Avg
		1	2	3	4	
UNISA(Ours)	-	**67.50±0.00**	**58.27±1.51**	**51.67±0.77**	**43.76±1.50**	**55.30±1.13**
UNISA(Ours)	Brightness	67.50±0.00	58.27±1.51	51.67±0.77	43.76±1.50	55.30±1.13
UNISA(Ours)	Contrass	67.50±0.00	58.27±1.51	51.67±0.77	43.76±1.50	55.30±1.13
UNISA(Ours)	Hue	67.50±0.00	58.27±1.51	51.67±0.77	43.76±1.50	55.30±1.13
UNISA(Ours)	Saturation	67.50±0.00	58.27±1.51	51.67±0.77	43.76±1.50	55.30±1.13
UNISA(Ours)	Blur	67.50±0.00	58.27±1.51	51.67±0.77	43.76±1.50	55.30±1.13
UNISA(Ours)	Gaussian	66.25±0.00	57.67±0.92	49.72±1.85	42.86±1.11	54.13±1.17
Baseline[15]	Gaussian	19.58±0.00	17.33±0.87	14.22±1.44	11.71±0.95	15.71±0.97
F2M[15]	Gaussian	25.42±0.00	16.27±0.39	13.39±1.27	10.81±1.13	16.47±0.87
FSLL[12]	Gaussian	24.17±0.00	11.73±1.29	9.39±1.26	7.90±1.01	13.30±1.03
ICARL[37]	Gaussian	26.00±2.55	20.40±0.50	17.28±1.39	14.43±0.86	19.53±1.53
Rebalance[70]	Gaussian	13.33±0.40	10.80±0.40	9.00±0.12	7.57±0.16	10.18±0.30
S3C[16]	Gaussian	53.75±0.00	41.40±1.79	33.11±0.56	30.14±1.45	39.60±1.19
SCN[2]	Gaussian	21.67±0.00	10.67±1.57	8.61±1.90	8.76±1.11	12.43±1.35
ORCO[51]	Gaussian	35.83±0.00	10.67±2.49	14.89±3.32	15.62±2.74	19.25±2.49

cannot be handled by the existing FSCL algorithms calling for fully labeled samples. There are no direct competitors of UNISA specifically devised for UFSCL problems. In addition, RS images are also difficult to deal with because of their spatiotemporal natures. It is seen by low base task accuracy of consolidated algorithms except UNISA. On the other hand, UNISA copes with the RS data and low labels equally well where its base task accuracy is significantly better than prior arts. Fig. 5 visualizes the trace of classification accuracy per task across all the four datasets. It is seen that UNISA is superior to other seven algorithms with large margins. Its performances are consistent across all the tasks and the accuracy of the base task is significantly higher than other algorithms. This fact indicates the success of UNISA in handling low-label situations and spatiotemporal nature of RS images.

C. Ablation Study

This section analyzes UNISA performances under five different configurations where numerical results are reported in Table IV. Our ablation study is run under the UCMerced problem. Configuration A presents the absence of flat minima concepts, while configuration B portrays UNISA's performances without the wide minima concept. Configurations C and D, respectively, stand for the absence of prototype scattering loss and the positive alignment loss. Finally, configuration E offers the case of without the ball generator loss. It is confirmed that each learning configuration plays key roles

TABLE VII

ROBUSTNESS ANALYSIS OF THE PROPOSED METHOD IN VARIOUS HYPER-PARAMETERS SETTINGS ON THE FIVE-SHOT UCMERCED DATASET. MIN.LR AND BS DENOTE MINIMUM LEARNING RATE AND BATCH SIZE, RESPECTIVELY

Hyper Params.		Session				Avg
Min.LR	BS	1	2	3	4	
0.001	128	67.50±0.00	58.27±1.51	51.67±0.77	43.76±1.50	55.30±1.13
0.0001	128	67.50±0.00	58.27±1.51	51.67±0.77	43.76±1.50	55.30±1.13
0.00001	128	67.50±0.00	58.27±1.51	51.67±0.77	43.76±1.50	55.30±1.13
0.0	128	67.50±0.00	58.27±1.51	51.67±0.77	43.76±1.50	55.30±1.13
0.0	64	67.50±0.00	58.27±1.51	51.67±0.77	43.76±1.50	55.30±1.13
0.0	32	67.50±0.00	58.27±1.51	51.67±0.77	43.76±1.50	55.30±1.13
0.0	16	67.50±0.00	58.27±1.51	51.67±0.77	43.76±1.50	55.30±1.13
0.0	8	67.50±0.00	57.20±1.74	49.61±1.79	43.38±0.96	54.42±1.00
0.0	4	67.50±0.00	55.40±2.00	47.56±2.16	40.10±1.47	52.64±1.64

TABLE VIII

SENSITIVITY ANALYSIS OF THE PROPOSED WITH VARIOUS BOUND (B) VALUES ON THE FIVE-SHOT UCMERCED DATASET

Bound (b)	Session				Average
	1	2	3	4	
0.001	69.58±0.00	57.73±1.85	51.11±1.35	43.48±1.23	55.48±1.30
0.005	69.58±0.00	58.00±1.81	51.11±0.92	43.86±1.36	55.64±1.22
0.01	67.50±0.00	58.27±1.51	51.67±0.77	43.76±1.50	55.30±1.13
0.05	69.58±0.00	58.00±1.81	51.11±0.92	43.86±1.36	55.64±1.22
0.1	68.33±0.00	58.07±1.20	50.61±0.42	42.43±0.52	54.86±0.69

TABLE IX

IMPACT OF THE NUMBER OF PERTURBATION LOOP (M) ON THE PROPOSED METHOD PERFORMANCE ON THE FIVE-SHOT UCMERCED DATASET

Perturbation loop (M)	Session				Average
	1	2	3	4	
1	64.58±0.00	55.87±1.43	48.94±1.09	40.90±0.95	52.57±1.02
2	67.50±0.00	58.27±1.51	51.67±0.77	43.76±1.50	55.30±1.13
3	65.42±0.00	57.07±1.40	49.33±1.75	41.57±1.63	53.35±1.39
4	67.50±0.00	60.27±1.29	53.39±0.81	46.00±1.34	56.79±1.01
5	72.50±0.00	58.93±2.04	50.22±1.01	43.43±0.41	56.27±1.16
6	75.83±0.00	63.20±1.59	54.28±1.22	45.57±0.99	59.72±1.12
7	79.17±0.00	66.07±1.11	56.78±1.33	47.86±0.48	62.47±0.90

in UNISA's performances where the absence of one loss function leads to degraded performances. The most substantial deterioration is observed in configuration D without positive alignment loss where the performance is compromised by over 12%. This finding demonstrates the presence of class collision issue. The absence of prototype scattering loss in configuration C also leads to dramatic performance losses. This result signifies lacks of uniformity term bringing down generalization performances. Configurations A and B also exhibit lower performances than UNISA attributed to the CF problem. Configuration E is worse by about 3% than UNISA.

D. Balance of Loss Coefficients

We extend our ablation study to investigate the impact of loss coefficient $\lambda_i, i \in [1, 5]$ on average performance of UNISA. Table V shows the average performance of UNISA in variations in λ_i. The upper part of the table (U1–U8) shows the impact of λ_1, λ_2, and λ_3 that is applied since the base task, while the lower part of the table (U9–U11) shows the impact of λ_4 and λ_5 that is applied in few-shot tasks only. The table shows that the lower value of λ_1 costs the highest performance drop i.e., about 5%. While the lower value of λ_2 or λ_3 decreases the average performance by about 1%.

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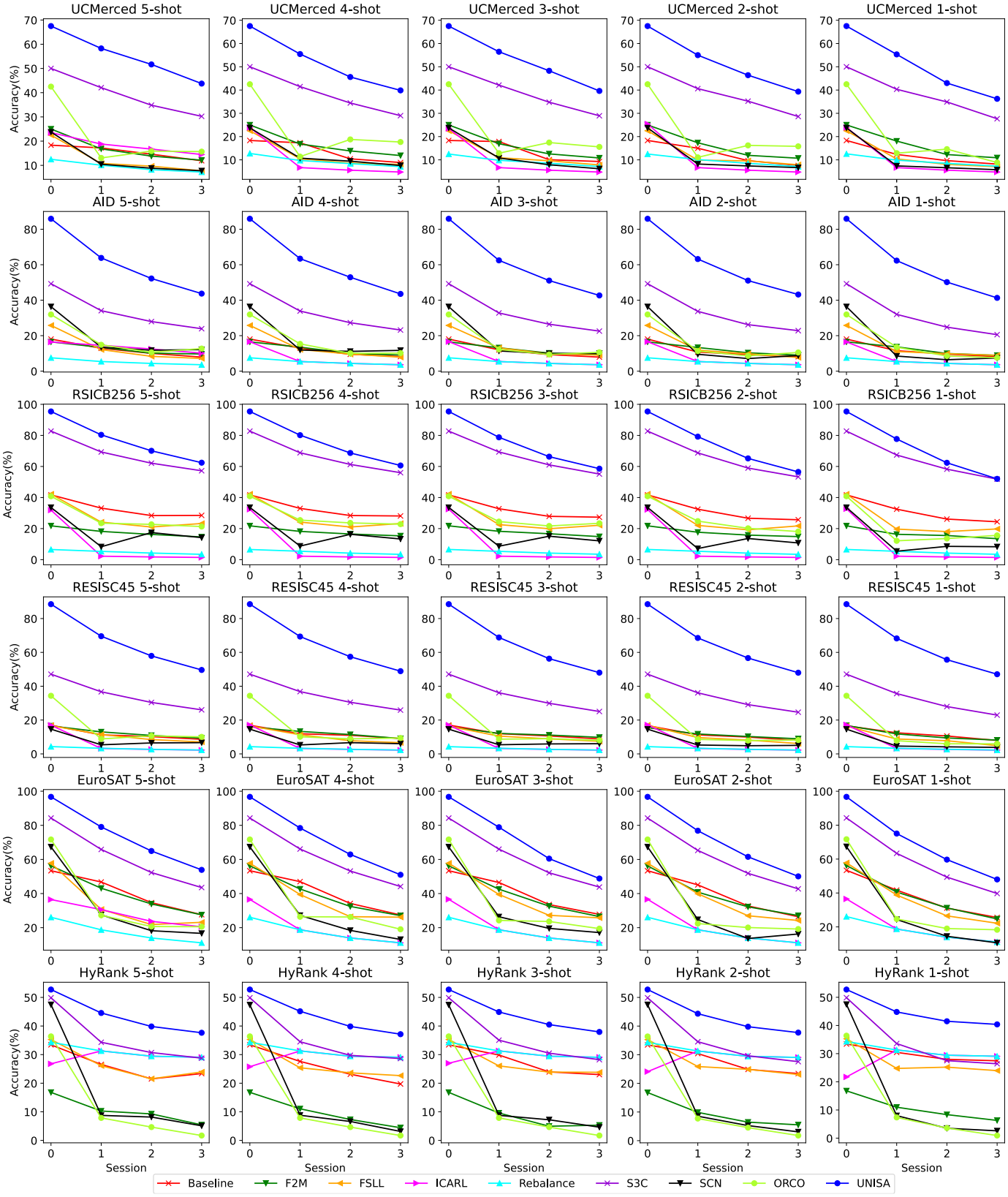


Fig. 5. Visualization of performances of consolidated algorithms across six datasets in one- to five-shot settings.

Note that the zero value of λ_1 and λ_2 (the absence of $\mathcal{L}_{\text{psa}}$ and $\mathcal{L}_{\text{cpr}}$) drops the average accuracy by 12% and 2%, respectively. It means that the lower coefficients still achieve a higher performance than zero. The table shows that the proposed method maintains its performance in a lower value of λ_4 or λ_5.

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TABLE X
RUNNING TIME OF THE CONSOLIDATED ALGORITHMS

Method	Training time per epoch (s)											
	UCMerced		AID		RSICB256		RESISC45		EuroSAT		HyRank	
	Base task	Few-shot task	Base task	Few-shot task	Base task	Few-shot task	Base task	Few-shot task	Base task	Few-shot task	Base task	Few-shot task
Baseline[15]	0.54±0.01	0.04±0.00	1.33±0.02	0.09±0.02	0.73±0.02	0.15±0.01	0.96±0.09	0.11±0.02	0.65±0.01	0.15±0.01	0.62±0.01	0.10±0.00
F2M[15]	0.64±0.01	0.73±0.05	1.38±0.06	1.11±0.08	1.54±0.01	0.57±0.03	1.13±0.06	0.41±0.02	0.84±0.01	0.68±0.05	0.65±0.00	0.78±0.02
FSLL[12]	0.54±0.01	0.68±0.07	1.33±0.02	1.36±0.04	0.73±0.02	1.02±0.05	0.96±0.09	0.36±0.05	0.65±0.01	0.7±0.08	0.62±0.01	0.67±0.02
ICARL[37]	0.59±0.02	0.30±0.01	1.27±0.04	0.62±0.10	0.71±0.03	1.64±0.16	0.67±0.07	1.77±0.15	0.79±0.10	1.37±0.06	0.60±0.00	1.35±0.04
Rebalance[70]	0.58±0.02	0.34±0.00	1.15±0.01	1.67±0.42	0.81±0.04	1.64±0.16	0.65±0.10	1.68±0.13	0.63±0.01	1.26±0.02	0.79±0.00	1.27±0.01
S3C[16]	1.18±0.05	0.62±0.01	3.14±0.04	0.95±0.06	2.11±0.35	0.84±0.08	2.90±0.05	1.28±0.02	2.41±0.72	0.86±0.02	1.51±0.09	0.76±0.02
SCN[2]	0.84±0.01	2.40±0.17	6.53±0.20	7.05±0.94	3.60±0.92	7.10±0.54	4.03±0.00	8.20±0.30	2.17±0.56	5.97±0.89	1.64±0.14	7.63±0.21
ORCO[51]	2.20±0.13	1.15±0.01	4.31±0.22	1.04±0.03	2.59±0.14	0.83±0.02	3.62±0.75	1.38±0.06	2.28±0.22	1.18±0.11	2.71±0.36	1.25±0.02
UNISA(Ours)	2.60±0.55	1.88±0.18	11.67±0.58	3.52±0.56	20.67±0.58	2.34±0.50	29.33±0.58	1.67±0.58	17.33±0.58	1.58±0.37	4.33±0.58	2.11±0.36

E. Robustness Analysis

This section discusses the robustness analysis of the proposed method against noise attacks on testing data and different hyperparameter settings.

1) Against Noises: Table VI shows that the proposed method consistently maintains its performance against brightness, contrast, hue, saturation, blur, and Gaussian [76] noises. The Gaussian noise attack reduces its performance by about 1% on average, while the other noise attacks do not impact its performance. In addition, the table shows that the proposed method consistently achieves the highest performance with a significant gap ($\geq$ 14%) compared with the competitor methods under the same noise attack i.e., Gaussian noise.

2) In Various Hyperparameter Settings: Table VII shows the robustness of our proposed method under different hyperparameter settings i.e., minimum learning rate and batch size. Note that our method uses a learning rate predictor by cosine annealing that takes the minimum learning rate as its lower bound. The table shows that the performance of our proposed method decreases only on the smallest batch size settings, i.e., 4 and 8.

F. Sensitivity Analysis

This section offers our sensitivity analysis done with the UCMerced problem. Specifically, different flat minima bounds b are attempted as well as the number of perturbation loops M. Tables VIII and IX report our numerical results. Different flat minima bounds do not result in significant differences in performances. Nevertheless, a too-large bound causes performance deterioration because of inaccurate approximations of flat-wide minima regions contributing to the CF problem. An interesting observation is found with respect to the perturbation loop M where the least performance is attained with $M = 1$ because there does not exist any flat-wide minima estimations. UNISA's performances constantly increase with the increase in the perturbation loop M, i.e., the highest performance is attained with $M = 7$ since this step results in fine-grained approximations of flat-wide minima regions. Note that extra perturbation loop M incurs additional computational burdens.

G. Complexity Analysis and Execution Times

See our supplemental document for a detailed complexity analysis. Our complexity analysis shows that our method has $O(N)$ complexity, where N is the size of the dataset accumulated from the first task until the last task. Table X reports the training time per epoch for all the consolidated algorithms. UNISA imposes higher execution times than other consolidated algorithms in the base task. The reason for this is found in the perturbation mechanism of UNISA where M perturbation loops are required to discover flat-wide minima regions. Note that other algorithms use 10% of base samples due to the absence of true class labels, thus expediting their running times. On the other side, the execution times of few-shot tasks of UNISA are comparable to other algorithms and always lower than SCN.

VII. LIMITATIONS

UNISA has demonstrated encouraging performances compared with prior arts while addressing an unexplored problem, UFSCL. Nonetheless, there exist some rooms for further studies.

1) Although the UFSCL problem is highly challenging and concerns on the problem of label scarcities in the FSCL, all the tasks are drawn from the same domain. This opens a future research issue to address the problem of cross-domain FSCL involving several domains.
2) Although UNISA is memory-free and unlike [15], it relies on the data augmentation procedure to address a sequence of few-shot learning tasks. Such mechanism simplifies the FSCL problem to some extents. Another research issue is how to devise an augmentation-free FSCL approach. Note that [11], [12], [14], [15] usually bias toward the base task notwithstanding that they do not apply any data augmentation procedures. That is, they are only capable of classifying the base classes while performing poorly for few-shot classes.

VIII. CONCLUSION

This article proposes an extension of FSCL, namely, UFSCL where no labels are exploited for inducing the training process including its solution, UNISA. UNISA is built upon the flat-wide learning approach to address the CF problem while the concept of prototype scattering and positive sampling is implemented as a representation learning strategy. The efficacy of UNISA has been validated in the RS image scene classification problems where it produces the most encouraging numerical results over its prior arts. Our future work is devoted to address the data privacy problem in CL.

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